An Introduction to Markov Chains

What is a Stochastic Process?

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- Think of it as a system that changes randomly.
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  - The number of customers in a store at each hour.
  - The weather (Sunny, Rainy, Cloudy) each day.

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- **The Challenge:** The future state of the system could depend on its entire past history, making it very complex to model.
- **Question:** Can we simplify this? What if the future only depends on the **present**, not the entire past?

The Markov Property: "Memorylessness"

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- Mathematically, for a discrete-time process X_n:

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- **In simple terms:** Knowing the present makes the past irrelevant for predicting the future.
- A process with this property is called a **Markov Chain**.

Transition Probabilities and the Transition Matrix

- If a Markov chain is **time-homogeneous**, the probability of moving from one state to another does not change over time.
- We can define the one-step **transition probability** as:

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- We can organize these probabilities into a **Transition Matrix**, P.

The Transition Matrix P

If a system has k states, the transition matrix is a $k \times k$ matrix where the entry in the i-th row and j-th column is p_{ij}.

$$P = \begin{pmatrix} p_{11} & p_{12} & \cdots & p_{1k} \\ p_{21} & p_{22} & \cdots & p_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ p_{k1} & p_{k2} & \cdots & p_{kk} \end{pmatrix}$$

Property: Every row must sum to 1, since from any state i, the system must transition to *some* state.

The Power of the Transition Matrix

The transition matrix P is more than just a table of probabilities; it's the engine for analyzing the entire Markov chain.

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- **Multi-Step Transitions:** The n-step transition matrix, $P^{(n)}$, is simply P^n. The entry $(P^n)_{ij}$ gives the probability of moving from state i to state j in exactly n steps.
- **Long-Term (Steady-State) Analysis:** It is the key to finding the stationary distribution α by solving the fundamental equation:

$$\alpha P = \alpha$$

Problem 1: A Simple Weather Model

Problem

The weather in a town can be either Sunny (S) or Rainy (R).

- If it is Sunny today, there is an 80% chance it will be Sunny tomorrow.
- If it is Rainy today, there is a 60% chance it will be Rainy tomorrow.

If it is Sunny today, what is the probability it will be Sunny two days from now?

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If it is Sunny today, what is the probability it will be Sunny two days from now?

- **States:** $\{1\text{: Sunny, } 2\text{: Rainy}\}$

Problem 1: Weather Model Solution

Step 1: Build the Transition Matrix P

From the problem description:

- $p_{11} = P(S \to S) = 0.8 \implies p_{12} = P(S \to R) = 1 - 0.8 = 0.2$
- $p_{22} = P(R \to R) = 0.6 \implies p_{21} = P(R \to S) = 1 - 0.6 = 0.4$

$$P = \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{pmatrix}$$

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Step 2: Calculate the 2-Step Transition Matrix $P^{(2)}$

The probability of moving from state i to j in 2 steps is given by the matrix P^2.

$$\begin{aligned} P^2 = P \times P &= \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{pmatrix} \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{pmatrix} \\ &= \begin{pmatrix} (0.8)(0.8) + (0.2)(0.4) & (0.8)(0.2) + (0.2)(0.6) \\ (0.4)(0.8) + (0.6)(0.4) & (0.4)(0.2) + (0.6)(0.6) \end{pmatrix} \\ &= \begin{pmatrix} 0.72 & 0.28 \\ 0.56 & 0.44 \end{pmatrix} \end{aligned}$$

Step 3: Find the Answer

The probability of being Sunny in two days given it is Sunny today is the p_{11} entry of P^2, which is **0.72** or 72%.

Long-Term Behavior: Steady-State Probabilities

- **Question:** After the process runs for a very long time ($n \to \infty$), what is the probability of being in any given state?

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- This collection of probabilities $\alpha = (\alpha_1, \alpha_2, ..., \alpha_k)$ is called the **steady-state** or **stationary distribution**.

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Finding the Stationary Distribution α

If the distribution is stationary, then after one more step, it should remain the same. This gives us the fundamental equation:

$$\alpha P = \alpha$$

We solve this system of linear equations along with the constraint that the probabilities must sum to 1:

$$\sum_{j=1}^{k} \alpha_j = 1$$

Problem 2: Brand Loyalty

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A study of customer brand loyalty finds that 80% of people who buy Brand A will buy it again next time, while 20% switch to Brand B. Of those who buy Brand B, 30% will switch to Brand A next time, while 70% will remain loyal to Brand B. In the long run, what is the market share of each brand?

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- This is asking for the steady-state distribution.
- **States:** $\{1\text{: Buys Brand A, } 2\text{: Buys Brand B}\}$

Problem 2: Brand Loyalty Solution

Step 1: Define the Transition Matrix P

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Step 2: Set up the Steady-State Equations

Let the stationary distribution be $\alpha = (\alpha_A, \alpha_B)$. The equations are:

1. $\alpha P = \alpha \implies (\alpha_A, \alpha_B) \begin{pmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{pmatrix} = (\alpha_A, \alpha_B)$
2. $\alpha_A + \alpha_B = 1$

Step 3: Solve the System of Equations

From $\alpha P = \alpha$, we get two equations:

- $0.8\alpha_A + 0.3\alpha_B = \alpha_A \implies -0.2\alpha_A + 0.3\alpha_B = 0$
- $0.2\alpha_A + 0.7\alpha_B = \alpha_B \implies 0.2\alpha_A - 0.3\alpha_B = 0$ (Note: this is the same eq.)

So we solve:

$$\begin{aligned} -0.2\alpha_A + 0.3\alpha_B &= 0 \\ \alpha_A + \alpha_B &= 1 \implies \alpha_B = 1 - \alpha_A \end{aligned}$$

Substitute the second into the first:

$$\begin{aligned} -0.2\alpha_A + 0.3(1 - \alpha_A) &= 0 \\ -0.2\alpha_A + 0.3 - 0.3\alpha_A &= 0 \\ 0.3 &= 0.5\alpha_A \implies \alpha_A = 0.6 \end{aligned}$$

Therefore, $\alpha_B = 1 - 0.6 = 0.4$.

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Conclusion

In the long run, Brand A will have 60% of the market share and Brand B will have 40%.

Summary

Key Concepts

- A **Markov Chain** is a stochastic process where the future depends only on the present (the Markov Property).
- The **Transition Matrix** P contains the probabilities p_{ij} of moving from state i to state j in one step.
- The **n-step transition matrix** is found by computing P^n.
- The **Stationary Distribution** α describes the long-term behavior of the chain and can be found by solving $\alpha P = \alpha$ with $\sum \alpha_j = 1$.

- Markov chains are a powerful tool for modeling systems in finance, biology, engineering, and many other fields.